

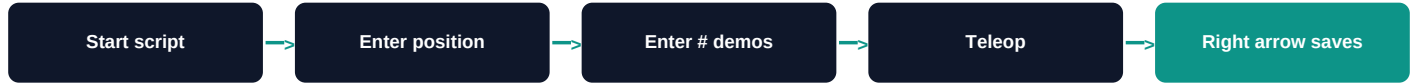
SO-101 Pick-and-Place Dataset Recording Guide

How to record teleoperated demonstrations · LeRobot v3.0 · Windows

1. Purpose

This guide explains how to record pick-and-place demonstrations on the SO-101 (leader + follower arms, dual cameras). The operator moves the **leader**; the **follower** mirrors. Each demo picks a cube from a numbered sheet position (1–50) and places it in a **fixed box**. Position and demo count are entered interactively. Spoken audio cues announce each phase.

High-level flow



2. Hardware defaults

Follower (robot)	COM3 · id my_so_arm · calibration/robots/so_follower
Leader (teleop)	COM4 · id my_so_arm · calibration/teleoperators/so_leader
Cameras	gripper_cam index 0 · top_cam index 1 · 640×480 @ 30 fps
Drop target	Fixed box (no position id)
Pickup id	Sheet positions 1–50 (entered per recording block)

3. Where files are stored

Dataset root	hf_data/so101-pick-place-positions/
Dataset id	aakashv100/so101-pick-place-positions
Totals / schema	.../meta/info.json
Episode metadata	.../meta/episodes/
State & actions	.../data/
Videos	.../videos/observation.images.gripper_cam/ .../videos/observation.images.top_cam/
Position ↔ episode log	.../pickup_positions.jsonl
Readable summary	.../pickup_positions_summary.json
Scripts	scripts/record_pickplace.ps1 (launcher) scripts/record_pickplace.py (recorder)

If the dataset folder already exists with saved episodes, running the script again **appends** new demos. It does not overwrite previous data.

4. What each episode records

Joints	6-DoF state + actions @ 30 fps
Cameras	gripper_cam + top_cam video; live side-by-side preview during recording
Task label	Includes pickup position, e.g. "Pick up the cube from sheet position N and place it in the box"
Sidecar log	JSONL maps episode_index → pickup_position for later analysis

5. How to start a recording session

1	Power both arms; connect USB; place cube on the sheet mark; box fixed at drop site
2	Confirm calibrations and camera indices (see §2)
3	Open PowerShell in the repo root

4	Activate the environment, then launch the recorder (commands below)
<pre>cd \so101-lerobot .\.venv\Scripts\Activate.ps1 .\scripts\record_pickplace.ps1</pre>	

Optional flags: -Fresh (fail if dataset already exists) · -PushToHub (upload when session ends) · -RobotPort / -TeleopPort / -GripperCamIndex / -TopCamIndex to override defaults.

6. Interactive recording steps

1	Run the launcher	Connects hardware; opens camera preview
2	Press Enter when preview looks correct	Shows text prompts
3	Enter pickup position (1–50)	Tags this block with that position id
4	Enter number of iterations	Plans that many demos for this position
5	Teleop one pick → place	Records joints + both cameras; may speak an audio cue
6	Press Right arrow when done	Saves the episode and updates the position log
7	Reset cube on the same mark	Audio: “Reset the environment”
8	Press Right arrow to end reset	Starts next iteration until the count is done
9	Enter next position, or q to quit	New block, or finalize and exit

7. Audio cues and controls

Spoken prompts use Windows text-to-speech (LeRobot `log_say`) so the operator can watch the robot instead of the terminal. Pass `--no-play-sounds` to disable.

“Position N, iteration i of M”	Next demo in the current block is starting
“Reset the environment”	Put the cube back; then Right arrow
“Re-record episode”	Last attempt discarded (after Left arrow)
“Stop recording” / “Exiting”	Session is ending

Terminal: position	Sheet pickup id for this block
Terminal: iterations	How many demos to record now
Terminal: Enter (blank)	Reuse last position or iteration count when offered
Terminal: q	Quit at a prompt
Right arrow	Save episode / end reset
Left arrow	Discard current episode and re-record
Esc	Stop the entire session

8. Continuing in a later session

Add new sheet positions	Run <code>.\scripts\record_pickplace.ps1</code> again; enter new position + iteration count
Add more demos for an existing position	Same command; enter that position and how many more
Start a brand-new dataset	Delete <code>hf_data/so101-pick-place-positions</code> , then run the launcher again

9. After recording (training)

The dataset is LeRobot-compatible (state, action, dual-camera video). Point ACT training at the local root, for example:

```
.\scripts\train_act.ps1 -RepoId aakashv100/so101-pick-place-positions -DatasetRoot  
hf_data/so101-pick-place-positions -JobName act_so101_pick_place_positions
```